

# Survey of Public Policies - Pre-Analysis Plan

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## Study Information

### Description

Does the political balance affect public policy approval? In this study, we aim to analyze factors beyond people's preferences of public policies. Specifically, we will operationalize "institutionalization" and "ownership" of policies to demonstrate how the perceived characteristics of public policies can increase or decrease public support, particularly among individuals with differing political views. This study is part of a master's thesis at the Brazilian School of Public and Business Administration (FGV-EBAPE).

This document outlines the design, expected results, and analysis plan of a conjoint survey experiment approved by the Committee for Ethical Compliance in Research Involving Human Beings at the Getúlio Vargas Foundation (FGV). The survey will ask participants about their political alignment with a party and the country's former presidents, their perceived level of institutionalization of certain federal public policies, their evaluation of these policies, and the association they perceive (or not) between policies and specific politicians.

We expect the data analysis to reveal: i) a positive association between individuals' positive partisanship and the evaluation of policies they perceive as aligned with their preferred parties; ii) a negative association between individuals' negative partisanship and the evaluation of policies they perceive as aligned with their least-favored parties; iii) a positive association between perceived institutionalization of the policies and their evaluation.

## Hypotheses

- Hypothesis 1: Institutionalization is not associated with policy ownership.
- Hypothesis 2a: Aligned policy ownership positively affects partisans' policy preferences.
- Hypothesis 2b: Misaligned policy ownership negatively affects anti-partisans' policy preferences.
- Hypothesis 2c: Policy ownership negatively affects non-partisans' policy preferences.
- Hypothesis 3: Institutionalization positively affects the individuals' policy preferences.

## Design Plan

### Main Variables

We will use a conjoint experiment to measure the most relevant features of institutionalization. Through randomization, participants will be shown fictional policies combining these features, and they will be asked

about their confidence in the policy's continuity. Based on the features' relevance predicted by the conjoint experiment, we will then ask participants which of these features they perceive in real policies to determine their level of institutionalization. Policies' ownership will be assessed by examining whether participants associate the policies with specific politicians, notably Lula (PT) and Bolsonaro (PL). To measure aligned policies, we will check if the association between policies and politicians matches the participants' preferred parties, if they have one. Finally, participants will be asked to evaluate a sample of policies. The study will be conducted with a cross-sectional sample of Brazilian internet users, using Qualtrics software for data collection and R for data analysis.

## Conjoint Experiment

We will employ a block randomization technique within the Qualtrics platform. Each participant will be randomly assigned to one of four fictional policies and will see two versions of it:

Por favor, leia o texto a seguir antes de responder as próximas perguntas:

1. [1/4] Com o aumento de casos de depressão e suicídios entre os jovens brasileiros, uma ONG recomendou que o país criasse uma política pública para melhorar a saúde mental entre os alunos do ensino básico. Uma opção seria o programa "Saúde Mental nas Escolas", que visa promover o bem-estar mental dos alunos treinando os professores para identificarem tendências depressivas e suicidas entre jovens e os acolherem. Nas duas páginas seguintes, você verá dois contextos diferentes em que o "Saúde Mental nas Escolas" pode ser aplicado e responderá em qual deles acredita que ele teria mais chances de continuar existindo.
2. [2/4] Diante do aumento da desigualdade de renda e da exclusão social em áreas urbanas, uma ONG recomendou que o governo implementasse uma política de moradia social para famílias de baixa renda. Uma opção seria o programa "Minha Casa Melhor", que promoveria a reforma de habitações populares em áreas de risco, com acesso a serviços básicos e infraestrutura adequada. Nas duas páginas seguintes, você verá dois contextos diferentes em que o "Minha Casa Melhor" pode ser aplicado e responderá em qual deles acredita que ele teria mais chances de continuar existindo.
3. [3/4] Considerando os impactos à saúde pública e a crescente popularidade dos cigarros eletrônicos no país, uma ONG recomendou que o Brasil criasse políticas de regularização do seu uso. Uma opção seria o programa "Eletrônico também é Cigarro!", que exigiria que as empresas colocassem avisos explícitos sobre os riscos à saúde nas embalagens de cigarros eletrônicos. Nas duas páginas seguintes, você verá dois contextos diferentes em que o "Eletrônico também é Cigarro!" pode ser aplicado e responderá em qual deles acredita que ele teria mais chances de continuar existindo.
4. [4/4] Um estudo recente mostrou que mais da metade dos homicídios no Brasil não são resolvidos. Por isso, uma ONG recomendou que o país criasse políticas de incentivo para o esclarecimento dessas mortes. Uma opção seria o programa "Índice Nacional de Esclarecimento de Homicídios", que poderia servir como incentivo para que os órgãos de segurança do país inteiro trabalhassem juntos para melhorar a resolução de casos não resolvidos. Nas duas páginas seguintes, você verá dois contextos diferentes em que o "Índice Nacional de Esclarecimento de Homicídios" pode ser aplicado e responderá em qual deles acredita que ele teria mais chances de continuar existindo.

Subsequently, one of these four policies will have two versions presented to participants, each of them mixing pre-determined institutionalization characteristics. Then, they will choose the alternative in which they have more confidence in the continuity.

- Em qual desses dois contextos de implementação você acredita que a política citada no texto anterior teria mais chances de continuar existindo? [Clique em cima da opção que escolher].

With the features' relevance predicted by the conjoint experiment, we proceed to ask which of these features participants see in real policies, finding their institutionalization level.

- Agora, gostaria de entender melhor sua visão a respeito dessas 8 (oito) políticas públicas. Por favor, marque as alternativas que você acredita que descrevem melhor as seguintes políticas [não há limite de opções, marque quantas alternativas achar que fazem sentido, na sua opinião]:

|                                         | Os principais partidos e políticos apoiam essa política | Essa política está na Constituição | Essa política existe há muito tempo | É gerida por funcionários públicos de carreira | Essa política costuma ter bons resultados | Não conheço a política. |
|-----------------------------------------|---------------------------------------------------------|------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------|-------------------------|
| Programa Bolsa-Família (PBF)            |                                                         |                                    |                                     |                                                |                                           |                         |
| Lei Rouanet                             |                                                         |                                    |                                     |                                                |                                           |                         |
| Governo Digital (gov.br)                |                                                         |                                    |                                     |                                                |                                           |                         |
| Escolas Cívico-Militares                |                                                         |                                    |                                     |                                                |                                           |                         |
| Benefício de Prestação Continuada (BPC) |                                                         |                                    |                                     |                                                |                                           |                         |
| Médicos Pelo Brasil                     |                                                         |                                    |                                     |                                                |                                           |                         |
| Random policy from the list             |                                                         |                                    |                                     |                                                |                                           |                         |
| Random policy from the list             |                                                         |                                    |                                     |                                                |                                           |                         |

- Two of the following policies will be shown for participants randomly: a) FIES/PROUNI; b) PAC (Programa de Aceleração do Crescimento); c) Lei de Cotas; d) PIX; e) ENEM (Exame Nacional do Ensino Médio); f) Minha Casa Minha Vida; g) Farmácia Popular; h) Mais Médicos; i) Pacote Anti-Crime; j) Reforma da Previdência; k) Casa Verde e Amarela; l) SAMU-192; m) SUS; n) MEI (Microempreendedor Individual); o) Teto de Gastos.

## Control Variables

In addition to the main variables, we intend to use information regarding age, gender, education, level of interest in politics, and income as control variables. This can also be useful to explore heterogeneities regarding the predicted values obtained in the joint experiment.

## Power Analysis

We will use the package “cjpwr” to conduct a power analysis for the conjoint and calculate the minimum sample size. For the survey, we will conduct the power analysis using the “pwr” package.

## Conjoint

```
library(cjpowR)

# This gives the minimum required effective sample size (type S, E(type M)):
df = cjpowr_amce(amce = 0.05, power = 0.8, levels = 2)

# For a conjoint with 2 profiles and 2 tasks, n becomes:
df$n/(2*2) # 22 answers = 88 effective answers
```

```
## [1] 782.9258
```

```
df$n/(2*3) # 508 answers = 3048 effective answers
```

```
## [1] 521.9505
```

```
#This gives the power (type S, E(type M)):
cjpowr_amce(amce = 0.05, n = 3132, levels = 2)
```

```
##           power      type_s exp_typeM amce      n alpha levels delta0
## 1 0.8000381 1.199855e-06 1.123466 0.05 3132 0.05      2 0.475
```

The required sample size for the conjoint is 3,132 effective answers, collected from 530 answers (22 with 2 task and 508 with 3 tasks). We expanded the tasks during the survey application because we changed the collection method, starting to recruit participants through a specialized company to overcome the difficulty of obtaining survey responses through Facebook ads during the Brazilian election period.

## Survey

```
library(pwr)
```

```
## Warning: package 'pwr' was built under R version 4.3.3
```

```
# Setting the parameters
r <- 0.2 # Tamanho de efeito equivalente à correlação esperada
alpha <- 0.05 # Nível de significância
power <- 0.90 # Poder desejado

# Calculating the sample size
result <- pwr.r.test(r = r, sig.level = alpha, power = power, alternative = "two.sided")

# Printing the result
print(result$n)
```

```
## [1] 257.756
```

The survey requires a 258 sample size calculated to detect an effect size of 0.2 with 90% power and a 5% significance level.

# Preparing Data

## Setting Libraries

## Importing Data

from the Survey through local archive

```
survey_1 <- read.csv("Thesis/Data/survey_1.csv") # importando dados
survey_2 <- read.csv("Thesis/Data/survey_2.csv")

# Juntando bases
survey <- bind_rows(survey_1, survey_2)

# Filtros e remoção das primeiras linhas
data_survey <- survey %>%
  slice(-c(1:2)) %>%
  filter(att_check == "Muito satisfeito(a)") # attention check

# Verificando tempo médio de duração
data_survey <- data_survey %>%
  mutate(
    Duration..in.seconds. = as.numeric(Duration..in.seconds.),
    Duration_in_minutes = Duration..in.seconds. / 60,
    duracao_max32 = pmin(Duration_in_minutes, 32)
  ) %>%
  select(1:6, Duration_in_minutes, duracao_max32, 7:ncol(data_survey))
```

from the Conjoint through local archive

```
conjoint_1 <- read.csv("Thesis/Data/conjoint_1.csv") # importando dados
conjoint_2 <- read.csv("Thesis/Data/conjoint_2.csv") # importando dados
conjoint_3 <- read.csv("Thesis/Data/conjoint_3.csv") # importando dados

# Filtros e remoção das primeiras linhas
data_conjoint_1 <- conjoint_1 %>%
  slice(-c(1:2)) %>%
  filter(att_check == "Muito satisfeito(a)") %>%
  filter(Finished == "True")

data_conjoint_2 <- conjoint_2 %>%
  slice(-c(1:2)) %>%
  filter(att_check == "Muito satisfeito(a)") %>%
  filter(Finished == "True")

data_conjoint_3 <- conjoint_3 %>%
  slice(-c(1:2)) %>%
  filter(att_check == "Muito satisfeito(a)") %>%
  filter(Finished == "True")
```

```

# Verificando tempo médio de duração do conjoint
data_conjoint_1 <- data_conjoint_1 %>%
  mutate(
    Duration..in.seconds. = as.numeric(Duration..in.seconds.),
    Duration_in_minutes = Duration..in.seconds. / 60
  ) %>%
  select(1:6, Duration_in_minutes, 7:ncol(data_conjoint_1))

data_conjoint_2 <- data_conjoint_2 %>%
  mutate(
    Duration..in.seconds. = as.numeric(Duration..in.seconds.),
    Duration_in_minutes = Duration..in.seconds. / 60
  ) %>%
  select(1:6, Duration_in_minutes, 7:ncol(data_conjoint_2))

data_conjoint_3 <- data_conjoint_3 %>%
  mutate(
    Duration..in.seconds. = as.numeric(Duration..in.seconds.),
    Duration_in_minutes = Duration..in.seconds. / 60
  ) %>%
  select(1:6, Duration_in_minutes, 7:ncol(data_conjoint_3))

```

## Adjusting conjoint data

Here, we renamed the variables and coded them to identify the profile the participant was viewing when answering the question.

```

# Mantendo só colunas do conjoint
conjoint_1 <- data_conjoint_1 %>%
  select(ResponseId, C1:DC22, feature0.1.1_CBCONJOINT:TextoConjoint_DO_text_red)

conjoint_2 <- data_conjoint_2 %>%
  select(ResponseId, C1:DC87, feature0.1.1_CBCONJOINT:TextoConjoint_DO_text_red)

conjoint_3 <- data_conjoint_3 %>%
  select(ResponseId, C1:C3, feature0.1.1_CBCONJOINT:TextoConjoint_DO_text_red)

# Juntando todas
conjoint <- bind_rows(conjoint_1, conjoint_2, conjoint_3)

# Selecionando C1 (1.1)
conjoint_c1.1 <- conjoint_1 %>%
  select(ResponseId, C1, DC21, ends_with("1.1_CBCONJOINT")) %>%
  rename(C = C1, DC = DC21,
    Apoio = feature0.1.1_CBCONJOINT,
    Legal = feature1.1.1_CBCONJOINT,
    Tempo = feature2.1.1_CBCONJOINT,
    Autonomia = feature3.1.1_CBCONJOINT,
    Resultados = feature4.1.1_CBCONJOINT) %>%
  mutate(out = ifelse(C == 1, 1, 0))

# Selecionando C1 (1.2)

```

```

conjoint_c1.2 <- conjoint_1 %>%
  select(ResponseId, C1, DC21, ends_with("1.2_CBCONJOINT")) %>%
  rename(C = C1, DC = DC21,
         Apoio = feature0.1.2_CBCONJOINT,
         Legal = feature1.1.2_CBCONJOINT,
         Tempo = feature2.1.2_CBCONJOINT,
         Autonomia = feature3.1.2_CBCONJOINT,
         Resultados = feature4.1.2_CBCONJOINT) %>%
  mutate(out = ifelse(C == 2, 1, 0))

# Seleccionando C2 (2.1)
conjoint_c2.1 <- conjoint_1 %>%
  select(ResponseId, C2, DC22, ends_with("2.1_CBCONJOINT")) %>%
  rename(C = C2, DC = DC22,
         Apoio = feature0.2.1_CBCONJOINT,
         Legal = feature1.2.1_CBCONJOINT,
         Tempo = feature2.2.1_CBCONJOINT,
         Autonomia = feature3.2.1_CBCONJOINT,
         Resultados = feature4.2.1_CBCONJOINT) %>%
  mutate(out = ifelse(C == 1, 1, 0))

# Seleccionando C2 (2.2)
conjoint_c2.2 <- conjoint_1 %>%
  select(ResponseId, C2, DC22, ends_with("2.2_CBCONJOINT")) %>%
  rename(C = C2, DC = DC22,
         Apoio = feature0.2.2_CBCONJOINT,
         Legal = feature1.2.2_CBCONJOINT,
         Tempo = feature2.2.2_CBCONJOINT,
         Autonomia = feature3.2.2_CBCONJOINT,
         Resultados = feature4.2.2_CBCONJOINT) %>%
  mutate(out = ifelse(C == 2, 1, 0))

# Seleccionando C3 (3.1)
conjoint_c3.1 <- conjoint_2 %>%
  select(ResponseId, C3, DC87, ends_with("3.1_CBCONJOINT")) %>%
  rename(C = C3, DC = DC87,
         Apoio = feature0.3.1_CBCONJOINT,
         Legal = feature1.3.1_CBCONJOINT,
         Tempo = feature2.3.1_CBCONJOINT,
         Autonomia = feature3.3.1_CBCONJOINT,
         Resultados = feature4.3.1_CBCONJOINT) %>%
  mutate(out = ifelse(C == 1, 1, 0))

# Seleccionando C3 (3.2)
conjoint_c3.2 <- conjoint_2 %>%
  select(ResponseId, C3, DC87, ends_with("3.2_CBCONJOINT")) %>%
  rename(C = C3, DC = DC87,
         Apoio = feature0.3.2_CBCONJOINT,
         Legal = feature1.3.2_CBCONJOINT,
         Tempo = feature2.3.2_CBCONJOINT,
         Autonomia = feature3.3.2_CBCONJOINT,
         Resultados = feature4.3.2_CBCONJOINT) %>%
  mutate(out = ifelse(C == 2, 1, 0))

```

```

# Empilhando os dataframes e reorganizando as colunas
conjoint_final <- bind_rows(conjoint_c1.1, conjoint_c1.2,
                           conjoint_c2.1, conjoint_c2.2,
                           conjoint_c3.1, conjoint_c3.2)

# Recodificando
conjoint_encoded <- conjoint_final %>%
  mutate(
    Apoio = factor(ifelse(Apoio == "Teve apoio dos principais políticos e partidos.", 1, 0)),
    Legal = factor(ifelse(Legal == "Está na Constituição.", 1, 0)),
    Tempo = factor(ifelse(Tempo == "Existe há muito tempo.", 1, 0)),
    Autonomia = factor(ifelse(Autonomia %in% c("É gerida por funcionários públicos de carreira.", "É ge
    Resultados = factor(ifelse(Resultados == "Conseguiu atingir os objetivos pretendidos pela política.
  )

```

## Creating the policy ownership variable (POLI\_ij)

The first independent variable will consist of a simple policy ownership indicator to check whether the respondent associated the program with a politician. Based on the “no - a little - yes” scale, we have a value from 0 to 2 for each politician the policy relates to. Adding the two together, we can have a policy ownership index from 0 to 4. We will also conduct correlation tests for this index with partisanship.

```

# Criando dataframe para variável
data_polit <- data_survey %>%
  select(ResponseId, ident_pt, ident_anti, starts_with("polit_")) %>% #selecionando colunas com polit_
  pivot_longer(cols = starts_with("polit_"),
              names_to = "policy",
              values_to = "politicization") %>% # pivotando
  mutate(politicization = as.integer(as.numeric(politicization))) %>%
  mutate(politicization = na_if(politicization, -99)) %>%
  mutate(policy = sub("^polit_", "", policy)) %>%
  separate(col = policy, into = c("policy", "policy_party"), sep = "_") %>%
  pivot_wider(names_from = policy_party, values_from = politicization) %>%
  rename(polit_pt = pt, polit_anti = anti) %>%
  mutate(polit_total = polit_pt + polit_anti) %>%
  mutate(
    concentration = case_when(
      polit_total == 0 ~ 0, # Evitar divisão por zero
      TRUE ~ pmax(polit_pt, polit_anti) / polit_total
    ) %>%
  mutate(POLI_ij = polit_total * concentration)

```

## Creating the aligned policy ownership variable (PART\_ij)

Secondly, we created the variable “aligned policy ownership” according to the respondent’s party identification. According to the most recent empirical findings, the PT and anti-PTismo delineate clear limits in Brazilian partisanship, so they will be our focus in the partisan analysis. When an interviewee is identified as petista or antipetista, we subtract the level of ownership associated with their party from the level of ownership associated with the rival party. On the other hand, in the case of non-partisans, we cannot define the value of this variable.



```
# Criando dataframe para variável
data_part <- data_polit %>%
  mutate(ident = case_when(
    ident_pt == "PT" ~ "petista",
    ident_anti == "PT" ~ "antipetista",
    TRUE ~ "nao_partidario")) %>%
  mutate(PART_ij = case_when(
    ident == "petista" ~ polit_pt - polit_anti,
    ident == "antipetista" ~ polit_anti - polit_pt,
    TRUE ~ NA_real_
  )) %>%
  select(ResponseId, policy, PART_ij)
```

## Creating the institutionalization variable (INST\_ij)

The institutionalization score is the result of the predicted value computed using the regression estimated in the conjoint and the responses of each individual to the policy attributes.

```
# Selecionar e renomear as colunas
data_inst <- data_survey %>%
  select(ResponseId, starts_with("inst_caract_"), -starts_with("inst_caract_D0")) %>%
  rename_with(~sub("inst_caract_", "", .))

# Criar novos dataframes
new_dataframes <- lapply(unique(sub("_[0-9]+$", "", names(data_inst)[-1])), function(prefix) {
  selected_cols <- grep(paste0("^", prefix, "_"), names(data_inst))
  new_data <- data_inst %>%
    select(ResponseId, starts_with(prefix)) %>%
    rename_all(~sub(paste0("^", prefix, "_"), "", .))
  new_data$policy <- prefix
  return(new_data)
})

# Empilhar todos os dataframes em um único dataframe
combined_data <- bind_rows(new_dataframes)

# Renomear as colunas
colnames(combined_data) <- c("ResponseId", "Apoio", "Legal", "Tempo", "Autonomia", "Resultados", "NS",

# Filtrar os casos
combined_data_filtered <- combined_data[rowSums(combined_data[, -c(1, ncol(combined_data))] != "") > 0,

# Filtrar os casos em que a coluna "NS" está preenchida e casos em que "Nenhuma" está preenchida junto
combined_data_filtered2 <- combined_data_filtered %>%
  filter(NS != "Não conheço a política") %>%
  filter(!(Nenhuma != "" &
    (Apoio != "" |
      Legal != "" |
      Tempo != "" |
      Autonomia != "" |
      Resultados != ""))) %>%
  select(-NS, -Nenhuma)
```

```

# Recodificando
data_inst <- combined_data_filtered2 %>%
  mutate(
    Apoio = factor(ifelse(Apoio == "Os principais partidos e políticos apoiam essa política", 1, 0)),
    Legal = factor(ifelse(Legal == "Essa política está na Constituição", 1, 0)),
    Tempo = factor(ifelse(Tempo == "Essa política existe há muito tempo", 1, 0)),
    Autonomia = factor(ifelse(Autonomia %in% c("É gerida por funcionários públicos de carreira.", "É ge
    Resultados = factor(ifelse(Resultados == "Essa política costuma ter bons resultados", 1, 0))
  )

# Logit regression
results_log <- glm(out ~ Apoio + Legal + Tempo + Autonomia + Resultados,
  data = conjoint_encoded, family = binomial(link = "logit"))

# summary(results_log)
# coef(results_log)

# Armazenando predict values
data_inst_predict <- predict(results_log, newdata = data_inst, type = "response")

# Levando o predict value para os dados
data_inst$inst_score <- data_inst_predict

# Criando índice de institucionalização das políticas
data_inst_index <- data_inst %>%
  group_by(policy) %>%
  summarize(INST_ij = mean(inst_score, na.rm = TRUE))

```

## Creating the evaluation variable (EVAL\_ij)

Finally, the evaluation will be carried out directly by the respondents, who will give ratings for each policy viewed by them.

```

# Criar tabela com valor de índice das políticas para conectar com dados da nota
index <- as.character(1:21)

name_policy <- c(
  "Programa Bolsa-Família (PBF)",
  "Benefício de Prestação Continuada (BPC)",
  "SAMU-192",
  "Mais Médicos",
  "Médicos pelo Brasil",
  "Farmácia Popular",
  "Casa Verde e Amarela",
  "Minha Casa Minha Vida",
  "FIES/PROUNI",
  "Escolas Cívico-Militares",
  "PAC (Programa de Aceleração do Crescimento)",
  "Lei Rouanet",
  "Lei de Cotas",
  "PIX",
  "Pacote Anti-Crime",

```

```

"MEI (Microempreendedor Individual)",
"ENEM (Exame Nacional do Ensino Médio)",
"Governo Digital (gov.br)",
"SUS",
"Reforma da Previdência",
"Teto de Gastos"
)
policy <- c(
  "pbf",
  "bpc",
  "samu",
  "maismed",
  "medpb",
  "farmpop",
  "casavam",
  "mcmv",
  "fiespro",
  "ecmilit",
  "pac",
  "rouanet",
  "cotas",
  "pix",
  "anticrime",
  "mei",
  "enem",
  "govbr",
  "sus",
  "prev",
  "teto"
)

policies <- data.frame(index, name_policy, policy)

# Criar o dataframe
data_eval <- data_survey %>%
  select(ResponseId, starts_with("nota_"), idade, sexo, regioao, interesse, escolar, renda) %>% #seleção
  pivot_longer(cols = starts_with("nota_"),
    names_to = "index",
    values_to = "EVAL_ij") %>% # pivotando
  mutate(index = sub("^nota_", "", index),
    EVAL_ij = as.numeric(EVAL_ij)) %>%
  left_join(policies) %>%
  select(ResponseId, policy, EVAL_ij, idade, sexo, regioao, interesse, escolar, renda)

## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'EVAL_ij = as.numeric(EVAL_ij)'.
## Caused by warning:
## ! NAs introduzidos por coerção

## Joining with 'by = join_by(index)'

```

## Joining all dataframes

```
data_final <- data_eval %>%
  left_join(data_polit, by = c("ResponseId", "policy")) %>%
  left_join(data_part, by = c("ResponseId", "policy")) %>%
  left_join(data_inst_index, by = c("policy")) %>%
  select(ResponseId, policy, concentration, INST_ij, POLI_ij, PART_ij, EVAL_ij, idade, sexo, regioao, in
```

## Results

### Expectations

We expect that the data analysis will reveal: i) no association between institutionalization and policy ownership; ii) a positive association between the evaluation of aligned owned policies and the preferences of partisan individuals; iii) a negative association between the evaluation of unaligned owned policies and the preferences of anti-partisan individuals; iv) a negative association between the evaluation of owned policies and the preferences of non-partisan individuals; v) a positive association between the perceived institutionalization of policies and their evaluation.

## Analyzing the Conjoint Experiment

```
# Renomear os casos
conjoint_amce <- conjoint_encoded %>%
  mutate(
    Apoio = factor(ifelse(Apoio == 1, "Apoio", "Não Apoio")),
    Legal = factor(ifelse(Legal == 1, "Legal", "Não Legal")),
    Tempo = factor(ifelse(Tempo == 1, "Tempo", "Não Tempo")),
    Autonomia = factor(ifelse(Autonomia == 1, "Autonomia", "Não Autonomia")),
    Resultados = factor(ifelse(Resultados == 1, "Resultados", "Não Resultados"))
  )

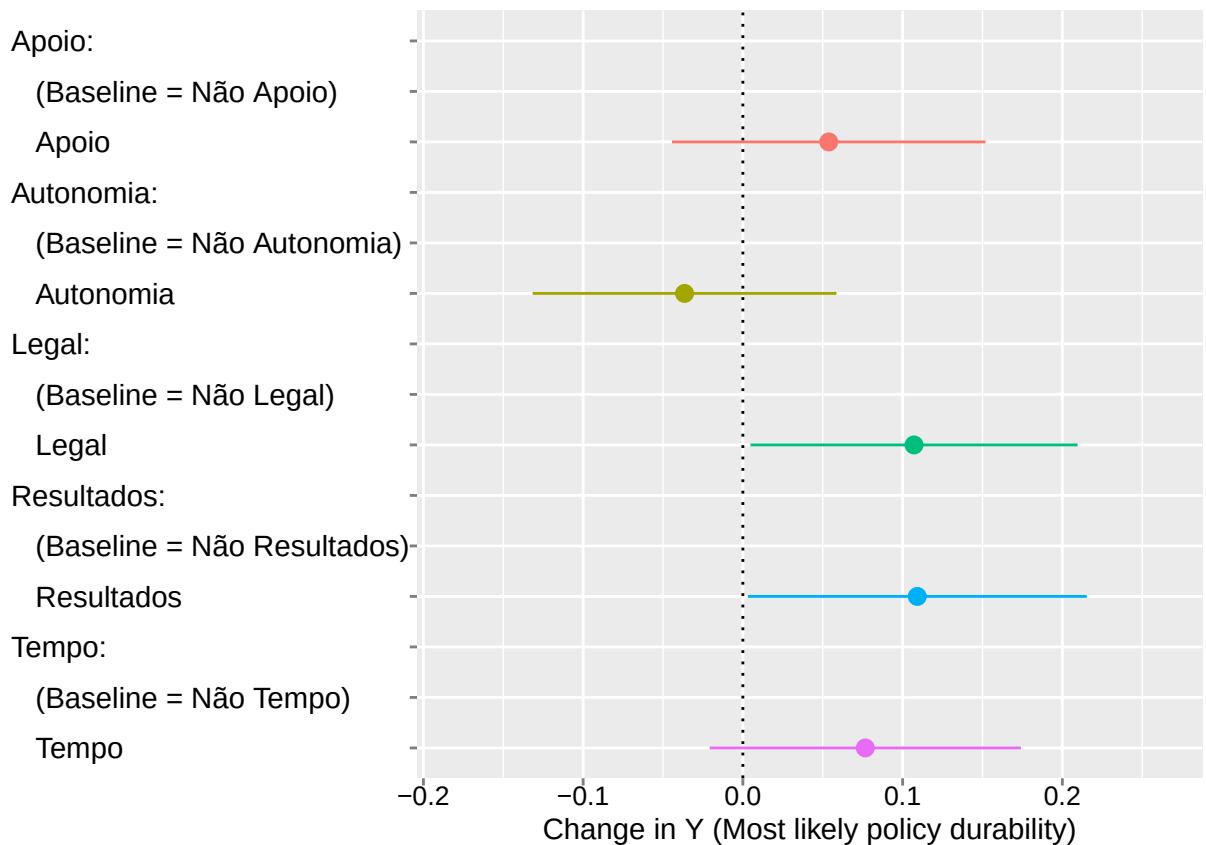
# Reordenar os níveis para garantir que os níveis com "Não" sejam o baseline
conjoint_amce <- conjoint_amce %>%
  mutate(
    Apoio = relevel(Apoio, ref = "Não Apoio"),
    Legal = relevel(Legal, ref = "Não Legal"),
    Tempo = relevel(Tempo, ref = "Não Tempo"),
    Autonomia = relevel(Autonomia, ref = "Não Autonomia"),
    Resultados = relevel(Resultados, ref = "Não Resultados")
  )

# Estimar os AMCEs
amce_results <- amce(
  data = conjoint_amce,
  formula = out ~ Apoio + Legal + Tempo + Autonomia + Resultados,
  respondent.id = "ResponseId"
)
```

```
# Mostrar os resultados dos AMCEs
summary(amce_results)
```

```
## -----
## Average Marginal Component Effects (AMCE):
## -----
##   Attribute      Level Estimate Std. Err  z value Pr(>|z|)
##   Apoio         Apoio  0.053775 0.050080  1.07377 0.282926
##   Autonomia     Autonomia -0.036500 0.048534 -0.75206 0.452016
##   Legal          Legal   0.107153 0.052246  2.05092 0.040275 *
##   Resultados    Resultados 0.109196 0.054142  2.01684 0.043712 *
##   Tempo         Tempo   0.076644 0.049684  1.54263 0.122921
## ---
## Number of Obs. = 480
## ---
## Number of Respondents = 223
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05
##
## -----
## AMCE Baseline Levels:
## -----
##   Attribute      Level
##   Apoio         Não Apoio
##   Autonomia     Não Autonomia
##   Legal          Não Legal
##   Resultados    Não Resultados
##   Tempo         Não Tempo
```

```
# Visualizar os AMCEs
plot(amce_results, xlab="Change in Y (Most likely policy durability)")
```



## Hypothesis 1

To test our hypothesis 1, that institutionalization is not associated with policy ownership, we will divide it into two stages. For an aggregate test, we will make a scatter plot of  $INST\_j$  x  $POLI\_j$ . These are the aggregated versions of our variables. While  $INST\_j$  is simply an average of  $INST\_ij$ ,  $POLI\_j$  is a measure that multiplies the level of  $POLI\_ij$  by the concentration of the policy's association with each politician. This concentration measure serves to distinguish a policy in which most participants clearly associate with a specific politician (more ownership) from one associated with both (less ownership, according to our theory). For the test that uses individuals as the unit of analysis, we compute the association between  $INST\_ij$  and  $PART\_ij$ . Furthermore, we will estimate a linear regression to also take into account some demographic controls, and cluster the standard errors by individual.

```
# Criar as variáveis agregadas
aggregated_data <- data_final %>%
  mutate(
    POLI_ij_scaled = POLI_ij / 4) %>%
  mutate(PART_ij_scaled = ((PART_ij + 2) / 4)) %>%
  group_by(policy) %>%
  summarise(
    mean_POLI_ij = mean(POLI_ij_scaled, na.rm = TRUE),
    mean_concentration = mean(concentration, na.rm = TRUE),
    POLI_j = mean_POLI_ij * mean_concentration,
    INST_j = mean(INST_ij, na.rm = TRUE),
    PART_j = mean(PART_ij_scaled, na.rm = TRUE),
    EVAL_j = mean(EVAL_ij, na.rm = TRUE),
```

```

    .groups = 'drop'
  )

# Scatter plot de INST_j x POLI_j
ggplot(aggregated_data, aes(x = POLI_j, y = INST_j, label = policy)) +
  geom_point() +
  labs(title = "Scatter Plot of INST_j vs POLI_j",
       x = "POLI_j",
       y = "INST_j") +
  geom_smooth(method = "lm", se = FALSE, color = "#F8766D") +
  geom_label_repel(box.padding = 0.5, point.padding = 0.3, max.overlaps = 10,
                  min.segment.length = 0.1,
                  color = "black") +
  theme_minimal()

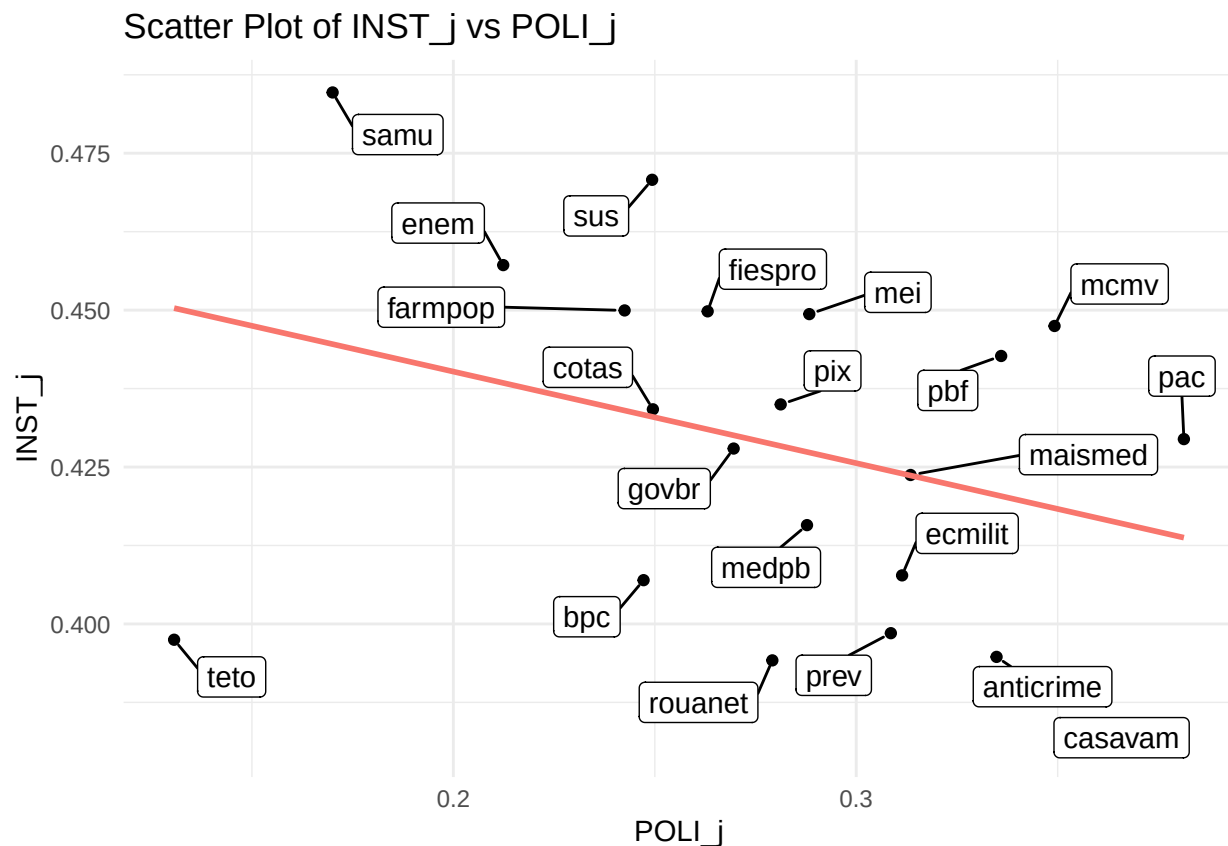
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```

## Warning: The following aesthetics were dropped during statistical transformation: label.
## i This can happen when ggplot fails to infer the correct grouping structure in
##   the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
##   variable into a factor?

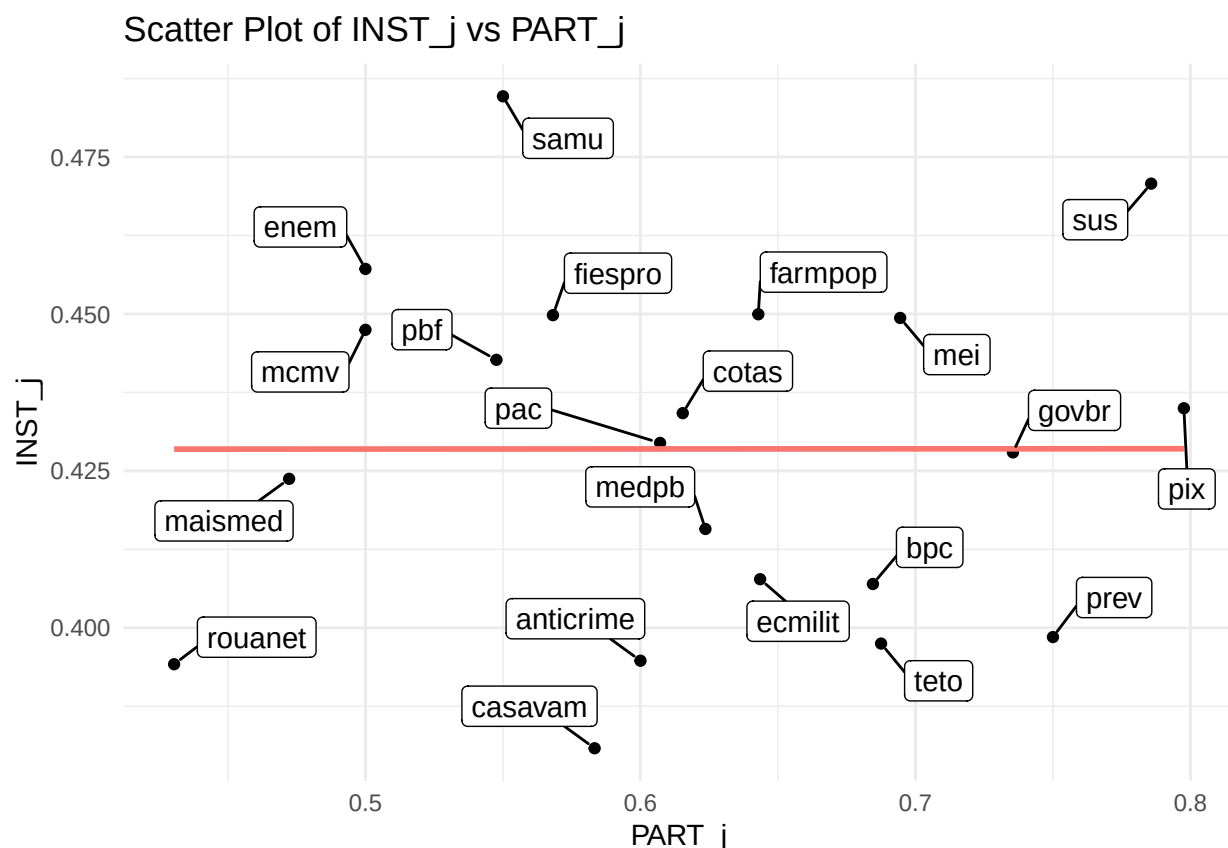
```



```
# Scatter plot de INST_j x PART_j
ggplot(aggregated_data, aes(x = PART_j, y = INST_j, label = policy)) +
  geom_point() +
  labs(title = "Scatter Plot of INST_j vs PART_j",
       x = "PART_j",
       y = "INST_j") +
  geom_smooth(method = "lm", se = FALSE, color = "#F8766D") +
  geom_label_repel(box.padding = 0.5, point.padding = 0.3, max.overlaps = 10,
                  min.segment.length = 0.1,
                  color = "black") +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation: label.
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
## variable into a factor?
```



```
# Correlações
cor.test(aggregated_data$INST_j, aggregated_data$POLI_j)
cor.test(aggregated_data$INST_j, aggregated_data$PART_j)
```



```

# Convertendo variáveis categóricas para fator e definindo o baseline (provisório)
data_final$sexo <- as.factor(data_final$sexo)
data_final$regiao <- as.factor(data_final$regiao)
data_final$interesse <- as.factor(data_final$interesse)
data_final$escolar <- as.factor(data_final$escolar)
data_final$renda <- as.factor(data_final$renda)

# Convertendo variáveis categóricas para fator e definindo o baseline (final)
#data_final$sexo <- relevel(as.factor(data_final$sexo), ref = "Feminino")
#data_final$regiao <- relevel(as.factor(data_final$regiao), ref = "Nordeste")
#data_final$interesse <- relevel(as.factor(data_final$interesse), ref = "Nada interessado(a)")
#data_final$escolar <- relevel(as.factor(data_final$escolar), ref = "Nunca frequentou escola")
#data_final$renda <- relevel(as.factor(data_final$renda), ref = "Nenhuma renda")

# Realizar a regressão linear com controles demográficos
model_1 <- lm(POLI_ij ~ INST_ij + idade + sexo + regiao + interesse + escolar + renda, data = data_final)
model_2 <- lm(PART_ij ~ INST_ij + idade + sexo + regiao + interesse + escolar + renda, data = data_final)

# Mostrar os resultados da regressão
summary(model_1)
summary(model_2)

# Ajustar os erros padrão para clustering por indivíduo
model_1_se <- coeftest(model_1, vcov = vcovCL, cluster = ~ResponseId)
model_2_se <- coeftest(model_2, vcov = vcovCL, cluster = ~ResponseId)

# Mostrar os resultados da regressão
summary(model_1_se)
summary(model_2_se)

# Imprimir os resultados da regressão
stargazer(model_1, model_2,
  title = "Regression Results",
  align = TRUE,
  no.space = TRUE,
  single.row = TRUE,
  out = "modelo_1_2.tex")

stargazer(model_1_se, model_2_se,
  title = "Regression Results",
  align = TRUE,
  no.space = TRUE,
  single.row = TRUE,
  out = "modelos_1_2_se.tex")

```

## Hypothesis 2a and 2b

For hypotheses 2a and 2b, which state that aligned policy ownership positively affects partisans' preferences and misaligned ownership negatively affects anti-partisans' preferences, we will estimate a regression model in which EVAL\_ij is the dependent variable and PART\_ij is the independent variable. We will estimate a model with and another without the control variables.

```

# Ajustando PART_ij
data_final <- data_final %>%
  mutate(PART_ij_scaled = ((PART_ij + 2)))

# Realizar a regressão linear com controles demográficos
model_3 <- lm(EVAL_ij ~ PART_ij_scaled + idade + sexo + regioao + interesse + escolar + renda, data = data_final)

# Ajustar os erros padrão para clustering por indivíduo
model_3_se <- coeftest(model_3, vcov = vcovCL, cluster = ~ResponseId)

# Visualizando resultados
summary(model_3)
summary(model_3_se)

# Imprimir os resultados da regressão
stargazer(model_3,
  title = "Regression Results for Model 3",
  align = TRUE,
  no.space = TRUE,
  single.row = TRUE,
  out = "modelo_3.tex")

stargazer(model_3_se,
  title = "Regression Results for Model 3",
  align = TRUE,
  no.space = TRUE,
  single.row = TRUE,
  out = "modelo_3_se.tex")

```

## Hypothesis 2c

As for hypothesis 2c, that policy ownership negatively affects non-partisans' policy preferences, EVAL\_ij remains the dependent variable, but POLI\_ij becomes the independent variable. This model, as predicted by the hypothesis, will be estimated only for non-partisans. Once again, versions with and without control will be estimated.

```

# Realizar a regressão linear com controles demográficos
model_4 <- lm(EVAL_ij ~ POLI_ij + idade + sexo + regioao + interesse + escolar + renda, data = data_final)

# Ajustar os erros padrão para clustering por indivíduo
model_4_se <- coeftest(model_4, vcov = vcovCL, cluster = ~ResponseId)

# Visualizando resultados
summary(model_4)
summary(model_4_se)

# Mostrar os resultados da regressão
stargazer(model_4,
  title = "Regression Results for EVAL_ij",
  align = TRUE,
  no.space = TRUE,
  single.row = TRUE,
  out = "modelo_4_.tex")

```

```
stargazer(model_4_se,
  title = "Regression Results for EVAL_ij",
  align = TRUE,
  no.space = TRUE,
  single.row = TRUE,
  out = "modelo_4_se.tex")
```

### Hypothesis 3

Finally, hypothesis 3 is that institutionalization positively affects individuals' policy preferences. Once again, EVAL\_ij is the dependent variable, while INST\_ij becomes the independent variable. This test will also have versions of the model with and without control variables.

```
# Realizar a regressão linear com controles demográficos
model_5 <- lm(EVAL_ij ~ INST_ij + idade + sexo + regioao + interesse + escolar + renda, data = data_final)

# Ajustar os erros padrão para clustering por indivíduo
model_5_se <- coeftest(model_5, vcov = vcovCL, cluster = ~ResponseId)

# Mostrar os resultados da regressão
stargazer(model_5,
  title = "INST_ij x EVAL_ij",
  align = TRUE,
  no.space = TRUE,
  single.row = TRUE,
  out = "modelo_5.tex")

stargazer(model_5_se,
  title = "INST_ij x EVAL_ij",
  align = TRUE,
  no.space = TRUE,
  single.row = TRUE,
  out = "modelo_5_se.tex")
```

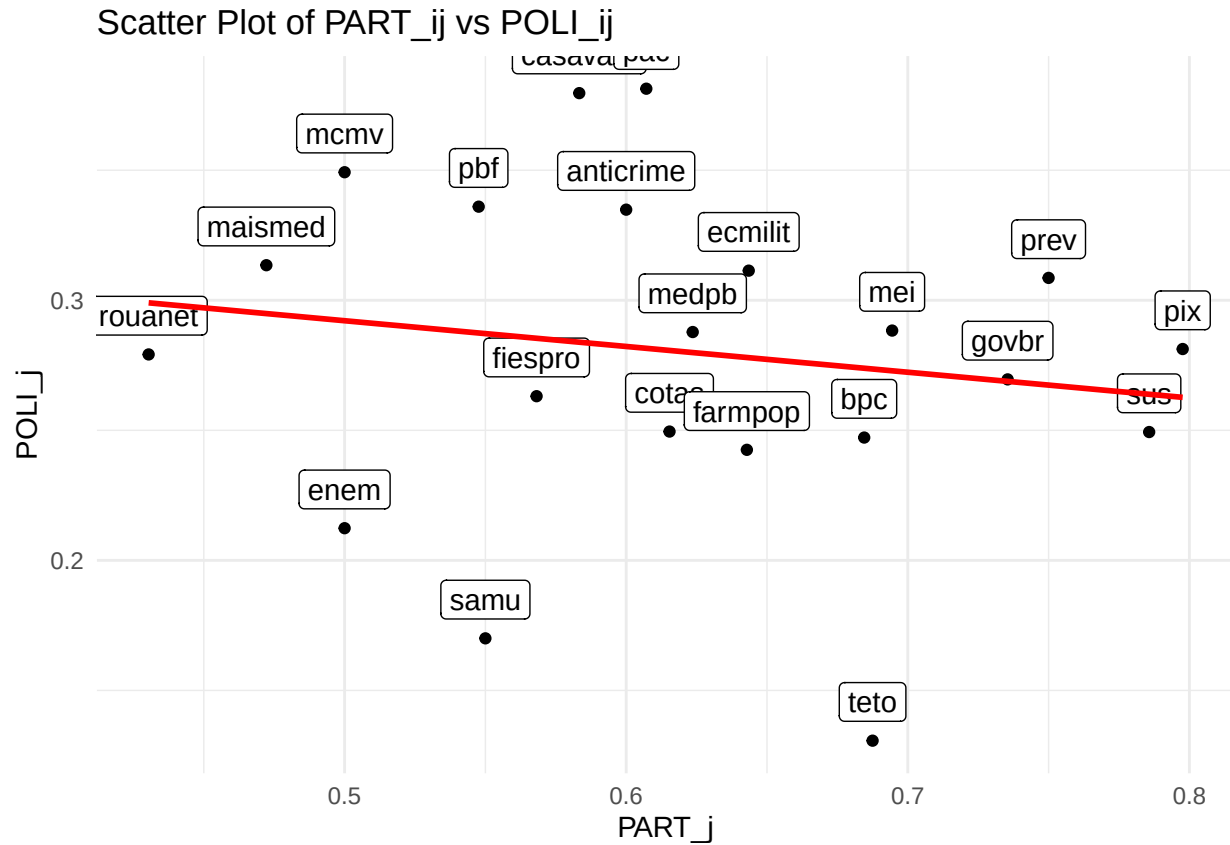
### Balance Checks

```
# Criar o gráfico de dispersão de POLI_ij x PART_ij
ggplot(aggregated_data, aes(x = PART_j, y = POLI_j, label = policy)) +
  geom_point() +
  geom_label(vjust = -0.5, hjust = 0.5) + # Ajustar a posição do texto
  labs(title = "Scatter Plot of PART_ij vs POLI_ij",
    x = "PART_j",
    y = "POLI_j") +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation: label.
```

```
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
## variable into a factor?
```



```
# Verificar a distribuição dos níveis de cada atributo específico
conjoint_encoded %>%
  count(Apoio) %>%
  mutate(percentage = n / sum(n) * 100)
```

```
##   Apoio    n percentage
## 1     0 255     53.125
## 2     1 225     46.875
```